

Joint convexity of the quantum relative entropy

Physics brief for `qRelativeEnt_joint_convexity` (PhysLean QuantumInfo)

The quantity

The classical relative entropy (Kullback–Leibler divergence) of two probability distributions,

$$D(P \parallel Q) = \sum_i p_i (\log p_i - \log q_i),$$

measures how efficiently data drawn from P can be told apart from data drawn from Q . Its quantum analogue, due to Umegaki, replaces distributions by density matrices:

$$D(\rho \parallel \sigma) = \text{Tr}[\rho (\log \rho - \log \sigma)],$$

with the standard conventions: natural logarithm (nats); logarithms taken on supports; and $D(\rho \parallel \sigma) = +\infty$ when $\text{supp } \rho \not\subseteq \text{supp } \sigma$ (in that case some event is possible under ρ but strictly impossible under σ , so the hypotheses can be separated at an unbounded rate). It is nonnegative and vanishes exactly when $\rho = \sigma$. Operationally, by the quantum Stein lemma, $D(\rho \parallel \sigma)$ is the optimal exponential rate at which the probability of mistaking ρ for σ decays in asymmetric hypothesis testing: it quantifies *distinguishability*.

What the theorem says

For states $\rho_1, \rho_2, \sigma_1, \sigma_2$ on a finite-dimensional space and $p \in [0, 1]$,

$$D(p\rho_1 + (1-p)\rho_2 \parallel p\sigma_1 + (1-p)\sigma_2) \leq p D(\rho_1 \parallel \sigma_1) + (1-p) D(\rho_2 \parallel \sigma_2).$$

Both arguments are mixed *simultaneously with the same weights*. This “joint” convexity is strictly stronger than convexity in each argument separately, and it is the jointness that carries the physical content.

The physical reading: a source flips a coin with bias p and prepares either ρ_1 or ρ_2 (the true scenario), or either σ_1 or σ_2 (the hypothesis). If the coin’s outcome is discarded, the two scenarios are described by the mixtures. The inequality says that forgetting the classical label can never make the two scenarios *more* distinguishable than they were, on average, with the label kept. Discarding information degrades distinguishability; it never creates it.

Why it matters

Joint convexity sits in a small cluster of statements at the analytic core of quantum information theory, each obtainable from the others by standard arguments:

- **Monotonicity / data processing.** Joint convexity yields monotonicity of D under partial trace, and, via Stinespring dilation, the full data-processing inequality $D(\Phi\rho\|\Phi\sigma) \leq D(\rho\|\sigma)$ for every quantum channel Φ (Lindblad, Uhlmann).
- **Strong subadditivity (SSA)** of von Neumann entropy (Lieb–Ruskai 1973) is essentially equivalent to it.
- **Second-law statements.** Non-negativity of entropy production for quantum dynamical semigroups relaxing to a stationary state follows from monotonicity of D (Spohn).
- **Resource theories and capacities.** Relative-entropy-based monotones (e.g. the relative entropy of entanglement) are monotones *because* of data processing, and the Holevo bound is a corollary as well.

Historically, the hard analytic input behind all of these is Lieb’s 1973 concavity theorem; everything above is downstream of some Lieb-type trace convexity. The accompanying literature note gives references and compares proof routes.

Where this sits in PhysLean

In the `QuantumInfo` library, states are `MState d` — positive-semidefinite Hermitian matrices of trace one over a finite index type `d` (`QuantumInfo/States/Mixed/MState.lean`) — and entropic quantities take values in $\mathbb{R}_{\geq 0\infty}$, so the infinite case is a value, not a side condition. The Umegaki entropy `D` is *defined* (`QuantumInfo/Entropy/Relative.lean`) as the $\alpha = 1$ member of the sandwiched Rényi family $\tilde{D}_\alpha$. The file `QuantumInfo/Entropy/DPI.lean` already contained substantial machinery for that family:

- the trace functional $\tilde{Q}_\alpha(\rho\|\sigma) = \text{Tr}[(\sigma^\gamma \rho \sigma^\gamma)^\alpha]$, $\gamma = (1 - \alpha)/(2\alpha)$, and its **joint convexity** for $\alpha > 1$ (`sandwichedTraceFunctional_jointly_convex`), proved by the Frank–Lieb variational argument on top of Lieb–Ando trace theorems formalized in `QuantumInfo/ForMathlib/HayataGroup/TraceInequality/`;
- **continuity** of $\alpha \mapsto \tilde{D}_\alpha(\rho\|\sigma)$ on $(0, \infty)$ (`sandwichedRelRentropy_continuousOn`), i.e. the $\alpha \rightarrow 1$ limit recovering the Umegaki entropy;
- the data-processing inequality for $\tilde{D}_\alpha$, $\alpha \geq 1$.

`qRelativeEnt_joint_convexity` itself had long been an open sorry (marked `@[sorryful]`) in `Relative.lean`. The present contribution closes it by an $\alpha \rightarrow 1^+$ limit over the existing machinery, in the same pattern by which the library already obtains the $\alpha = 1$ DPI from the $\alpha > 1$ case; the theorem now lives in `DPI.lean` because `Relative.lean` cannot import that file without a cycle. No new deep analytic input is introduced — the heavy lifting (Lieb-type concavity) was already formalized. What is new is the limit argument itself, including the degenerate mixing weights ($p \in \{0, 1\}$) and the infinite-value cases that the $\mathbb{R}_{\geq 0\infty}$ -valued statement forces one to treat explicitly.

Two statement-design points for readers coming from the physics literature: mixtures are written with the `Mixable` typeclass (`p [ρ1 ↔ ρ2]`), because `MState` is not an additive monoid and Mathlib’s `ConvexOn` therefore does not apply directly; and the theorem is stated for binary mixtures

(finite mixtures follow by the usual induction but are not separately formalized for $\mathbf{D}$, though the underlying $\tilde{Q}_\alpha$ lemma is already stated for arbitrary finite families). SSA is proved independently elsewhere in the library (`QuantumInfo/Entropy/SSA.lean`, via purification and weak monotonicity), so this theorem completes the convexity corner of the standard picture rather than serving as SSA’s source. The proof introduces no new axioms: `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`.

Grounding in the literature

Every physical claim above is standard and appears in the canonical references. This section names the credible sources so the statements are traceable; full bibliographic detail (with the arXiv-checked eprint corrections) is in `literature.qmd`. Lindblad 1974, Wilde 2017 (+ its arXiv precursor and licence), and the Carlen AMS listing were checked directly against Springer, Project Euclid, arXiv, and AMS on 2026-07-06; the remaining primary-source DOIs (Umegaki, Lieb 1973, Lieb–Ruskai 1973) are the standard published identifiers and are given for the reviewer to resolve.

Primary sources (the result itself).

- **Umegaki relative entropy** — H. Umegaki, *Conditional expectation in an operator algebra IV (entropy and information)*, Kōdai Math. Sem. Rep. **14** (1962), 59–85.
- **Joint convexity of the relative entropy** — G. Lindblad, *Expectations and entropy inequalities for finite quantum systems*, Commun. Math. Phys. **39** (1974), 111–119, DOI [10.1007/BF01608390](https://doi.org/10.1007/BF01608390). It follows from
- **Lieb’s concavity theorem** — E. H. Lieb, *Convex trace functions and the Wigner–Yanase–Dyson conjecture*, Adv. Math. **11** (1973), 267–288, DOI [10.1016/0001-8708\(73\)90011-X](https://doi.org/10.1016/0001-8708(73)90011-X).
- **Strong subadditivity (equivalent statement)** — E. H. Lieb, M. B. Ruskai, *Proof of the strong subadditivity of quantum-mechanical entropy*, J. Math. Phys. **14** (1973), 1938–1941, DOI [10.1063/1.1666274](https://doi.org/10.1063/1.1666274).

Textbook / course references (the physics context — openly available).

- **M. M. Wilde**, *Quantum Information Theory*, 2nd ed., Cambridge Univ. Press (2017), DOI [10.1017/9781316809976](https://doi.org/10.1017/9781316809976). Its open-access precursor is *From Classical to Quantum Shannon Theory*, [arXiv:1106.1445](https://arxiv.org/abs/1106.1445) (v8, 2019), **CC BY-NC-SA 3.0** — covers the relative entropy, its joint convexity, and the data-processing inequality, and is freely redistributable.
- **E. A. Carlen**, *Trace inequalities and quantum entropy: an introductory course*, in *Entropy and the Quantum*, Contemp. Math. **529**, Amer. Math. Soc. (2010), 73–140 — the self-contained mathematical route (Lieb concavity → joint convexity → SSA); [AMS listing](#).
- **M. Tomamichel**, *Quantum Information Processing with Finite Resources*, SpringerBriefs in Math. Phys. **5** (2016), [arXiv:1504.00233](https://arxiv.org/abs/1504.00233).
- **M. A. Nielsen, I. L. Chuang**, *Quantum Computation and Quantum Information*, 10th Anniv. ed., Cambridge Univ. Press (2010) — §11.3 (quantum relative entropy) for the standard graduate treatment.

A reviewer extract (openly licensed)

For the reviewer’s convenience, the standard textbook statement, as a paraphrase attributable to Wilde’s openly-licensed text ([arXiv:1106.1445](https://arxiv.org/abs/1106.1445), CC BY-NC-SA 3.0) and the equivalent treatments in Carlen and Nielsen–Chuang:

The quantum relative entropy $D(\rho\|\sigma)$ is jointly convex in its two arguments, and equivalently is monotone (non-increasing) under the action of any quantum channel — the data-processing inequality. This statement, the monotonicity of relative entropy, and the strong subadditivity of the von Neumann entropy are mutually derivable, and all rest on Lieb’s concavity theorem.

This is exactly the statement `qRelativeEnt_joint_convexity` formalizes (the joint-convexity half); the monotonicity half is the library’s separately-formalized DPI. No paywalled source is reproduced: the passage is a paraphrase grounded in the CC-licensed Wilde text and the open Carlen notes, with the primary sources (Lindblad, Lieb) cited by DOI only.